Christiaan Günter Alwyn Viviers

Applied ML Engineer & Research Scientist

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About Me

Postdoctoral researcher working on computer vision, multimodal foundation models, and generative modeling, with a focus on medical imaging and clinically-relevant AI. I am particularly interested in improving how machines learn from complex data, and in translating those learned representations into robust performance in high-stakes settings. Although grounded in healthcare, the methods I develop are broadly applicable across machine learning domains.

Technical Focus

- **Multimodal / Foundation Models:** self-supervised pretraining, vision-language alignment, representation learning.
- **Training at Scale:** data pipelines, reproducible experiments, benchmarking suites, distributed training, NVIDIA Run:ai
- **Optimization & Deployment:** quantization, throughput/latency profiling, batching/pipelining, memory optimization.
- **Generative Modeling:** diffusion / flow-based models, likelihood-free training, uncertainty quantification, image generation.
- **Edge Systems:** NVIDIA Jetson deployment, containerized inference services, parallel processing pipelines (multi-process / async).
- **Core Stack:** Python, PyTorch; Linux; Git; Docker; My everyday PC runs MS Windows.

Experience

Theta Vision — Jan 2025 – Present
Machine Learning Engineer — The Netherlands

- Led end-to-end ML lifecycle for real-world a endoscopic vision system: data ingestion/curation → training/finetuning → evaluation/robustness → ISO compliancy → deployment.
- Built and maintained the full engineering stack for NVIDIA Jetson deployments, including model quantization, hardware-aware profiling, and parallel processing pipelines for real-time throughput.
- Collaborated cross-functionally (engineers, clinicians, product stakeholders) to integrate model outputs into an operational prototype/workflow.
- Won the Purple NECtar X Innovation in Defense (PN x IID) 2025 challenge: how to optimally use synthetic data and generative models to detect novel objects.

Eindhoven University of Technology (TU/e) — Jun 2024 – Present
Postdoctoral Researcher — Image Generative Models (TASTI Project) — Eindhoven, The Netherlands

- Work package lead, facilitating the core of the project on tailoring image synthesis for industrial applications.
- Developed methods and **image generative models** to improve downstream utility of synthetic images across multiple sectors (healthcare, automotive, manufacturing, agriculture).
- Designed dataset curation and benchmarking pipelines to quantify synthetic-to-real transfer; emphasized reproducibility and robust evaluation.
- Collaborated with academic and industry partners; contributed to publications and technical dissemination.

Philips Image Guided Therapy Jun 2018 – Dec 2020
Software Engineer — Computer Vision The Netherlands

- Built a framework for integrated computer vision methods to support automated testing of production IGT systems - reducing time spent on manual verification tests.
- Worked with R&D and engineering teams on system-level validation, robustness, and maintainability of deployed vision solutions.

Stellenbosch University (SAND Microfluidics Lab) Jan 2016 – Mar 2016
Researcher — Biosensors & Microfluidics South Africa

- Developed and patented a biosensor prototype for bacteria detection; contributed to experimental design and prototyping.

Education

Eindhoven University of Technology & Philips 2020 – 2024
PhD, Electrical Engineering (Computer Vision & Signal Processing) The Netherlands
Thesis: *Enhanced Computer Vision Methods for Cancer Detection and Precision Guidance in Medical Imaging.*
Publications: CVPR, ICCV, ECCV, TIP, TMI, MICCAI (see Google Scholar.).

Stellenbosch University 2016 – 2017
MEng, Electrical Engineering (Biotechnology) South Africa
Thesis: *The Design and Fabrication of an Autophagy Flux Biosensor.* **Cum Laude**.
Patented: *Device for detecting target biomolecules*

Stellenbosch University 2012 – 2015
BEng, Electrical Engineering (Informatics) South Africa
Thesis: *Development of a Resistive Microfluidic Sensing Device for Pathogen Detection.*
Award: Jac Van der Merwe Prize (most innovative thesis in faculty).

Selected Publications (Foundation Models / Generative Modeling)

Scaling Self-Supervised and Cross-Modal Pretraining for Volumetric CT Transformers. *(CVPR 2026)*

Summary: Developed the largest Vision Language Model (VLM) for Computed Tomography.
Trained on a distributed NVIDIA DGX B200 system over multiple weeks. arXiv:2511.17209

Symmetrical Flow Matching: Unified Image Generation, Segmentation, and Classification with Score-Based Generative Models. *(AAAI 2026)*

Summary: Building on advances in diffusion models, we developed a generative Flow Matching matching model that unifies Image Generation, Segmentation, and Classification in a single model.
arXiv:2506.10634

Additional peer-reviewed publications: My list & Google Scholar.

Professional Services, Mentoring & Communication

- Experience in presenting technical work to mixed audiences (research + engineering + clinical/product stakeholders).
- Collaboration across academia and industry (TU/e, Philips, Theta Vision; multi-partner projects).
- Reviewer for ICCV, ECCV, MICCAI and various other computer vision conferences.
- Mentoring/supervision of MSc and PhD students.
- Taught Neural Networks for Computer Vision masters course for 5+ years at TU/e.

Languages

English (native), Afrikaans (native), Dutch (basic)